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Hydroxylated carbon nanotube enhanced sulfur cathodes for improved electrochemical performance of lithium–sulfur batteries†

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Hydroxylated multi-walled carbon nanotubes were introduced into sulfur cathodes to utilize the hydrophilic attraction between the OH group and polysulfides as well as to increase the utilization of sulfur. This approach shows a promising strategy for diminishing the polysulfides shuttling behavior and improving the electrochemical performance of lithium–sulfur batteries.

Battery-powered devices ranging from portable electronics to electric vehicles have greatly dominated our lives, but the development of lithium-ion batteries is gradually lagging behind the rapid development of those devices.[1] As a matter of fact, battery performance is largely dependent on galvanic electrode parts, especially the cathode component. As a promising cathode candidate, sulfur has received great interest owing to its high theoretical capacity of 1672 mA h g^{-1} and rich abundance.[1,2] Although lithium–sulfur batteries possess many merits as a promising energy storage solution, they still have several challenges that remain to be solved, including the intrinsic low electric conductivity of sulfur (5×10^{-30} S cm^{-1} at 25 °C) and the polysulfides shuttling effect.[3]

To address these challenges, many approaches based on carbon materials have been used to confine polysulfides physically in the cathode in order to increase their electrical conductivity and avoid polysulfides loss, including synthesizing the carbon–sulfur composite *via* sulfur melt adsorption[2,4] or inserting a carbon blocking layer in between the sulfur cathode and the separator.[5] Besides these approaches, chemical protection, such as use of hydroxyl functional groups (–OH) that can chemically bind with the highly polar polysulfides, has also proved to be a promising way to further stabilize polysulfides in carbon materials.[2,6]

In this communication, we present an effective method of introducing hydroxylated multi-walled carbon nanotubes (MWNTs) into the sulfur composite in order to improve the electrochemical performance of Li–S batteries. We prepared the sulfur–MWNT composite *via* an *in situ* sulfur deposition method, which is favorable to prepare the core–shell type of carbon–sulfur composites on a large scale.[7] Previous studies suggest that a hydrophilic functional group formed on carbon materials can effectively trap the polysulfide species due to their polarizing attraction.[2,8] Fig. 1 illustrates the hydrophilic behavior between polysulfides and OH-MWNTs. During the discharge process, lithium ions are electrochemically reacted with sulfur–MWNTs to form lithiated sulfur–MWNTs. In this step, the reduced polysulfides are dissolved out of the sulfur electrode and then attracted to the hydroxyl groups on the surface of the MWNTs by creating a chemical gradient, which efficiently inhibits the

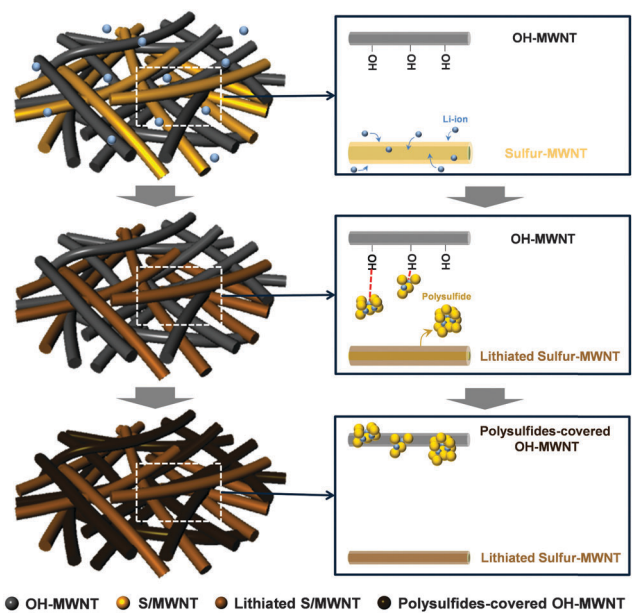

Fig. 1 Schematic illustration of the reaction between the OH-MWNTs and polysulfides during the discharge.

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diffusion of polysulfides out to the lithium metal and consequently alleviates the shuttle effect.

The scanning electron microscopy (SEM) images of the MWNTs and the as-prepared sulfur–MWNTs are presented in Fig. 2a and b, respectively. After depositing sulfur on the surface of the MWNTs, the sulfur–MWNT composite exhibits a core/shell structure and does not possess any agglomeration of sulfur clusters, indicating that the sulfur was uniformly coated onto the MWNTs (Fig. S1, ESI†). Transmission electron microscopy (TEM) was used to examine the structure of the sulfur–MWNT composite. As shown in the TEM images (Fig. 2c and d), highly crystallized MWNTs and conformal coated sulfur can be clearly observed. The thickness of sulfur coating was around 5 nm. Additionally, electron energy loss spectroscopy (EELS) mapping was carried out to investigate the distribution of sulfur along with the MWNTs (Fig. 2e). The EELS mapping shows that conformal sulfur coating was uniformly distributed on the MWNTs. X-ray powder diffraction (XRD) of the MWNTs, the sulfur–MWNT composite, and sublimed sulfur are presented in Fig. 2f. The sulfur–MWNTs and sublimed sulfur have a similar pattern with no shift of peaks, indicating that sulfur is well distributed on the surface of the MWNTs and the *in situ* sulfur deposition method does not affect the crystallinity of sulfur in the sulfur–MWNT composite.[9] For further analyzing the surface characteristic of OH-MWNTs and sulfur–MWNTs, XPS measurement was conducted and plotted in Fig. S2 (ESI†). The peak centered at 163.75 eV corresponds to the aroma peak of the C–S bond in sulfur–MWNTs, indicating that the C–S–C bond was formed between OH-MWNTs and sulfur by *in situ* sulfur deposition.[8c,10] To compare

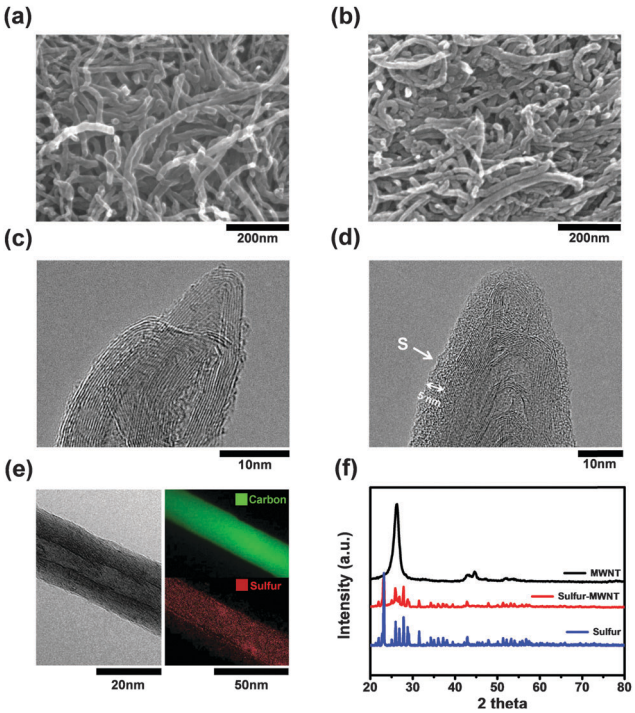

Fig. 2 SEM images of (a) MWNTs and (b) sulfur–MWNTs, TEM images of (c) MWNTs and (d) sulfur–MWNTs, (e) element mapping image using electron energy loss spectroscopy (EELS) for carbon and sulfur, and (f) XRD patterns.

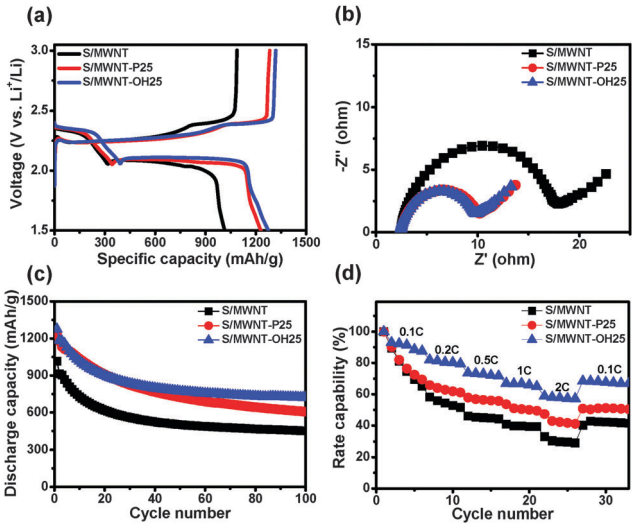

Fig. 3 Electrochemical characterizations of S/MWNT, S/MWNT-P25, and S/MWNT-OH25. (a) Initial discharge–charge curve, (b) Nyquist plot, (c) cycling performance, and (d) rate retention.

the effect of the OH group on the electrochemical performance of Li–S batteries, three types of samples including only sulfur–MWNTs (S/MWNTs), sulfur–MWNTs mixed with a 25 wt% pristine MWNTs (S/MWNT-P25), and sulfur–MWNTs mixed with 25 wt% OH-MWNTs (S/MWNT-OH25) were evaluated based on the sulfur mass, and the values assigned are 1.5 mg cm^{-2}, 1.18 mg cm^{-2}, and 1.19 mg cm^{-2}, respectively. Fig. 3a shows the increased initial discharge–charge capacity, when the MWNTs are introduced, which is due to the improved conductivity of the electrode.[11]

To analyse the electrochemical characteristics of each electrode, electrochemical impedance spectroscopy (EIS) measurements were conducted in the frequency range of 250 kHz to 10 mHz at an excitation amplitude of 10 mV. The Nyquist plots are composed of a semi-circle at high frequency and a linear line at low frequency (Fig. 3b). The intersection with the *x*-axis corresponds to Ohmic resistance between the electrode and electrolyte, and the values for the S/MWNT, S/MWNT-P25, and S/MWNT-OH25 are 2.5 Ω, 2.6 Ω and 2.5 Ω, respectively. On the other hand, the diameter of the semi-circle in the right part corresponds to the polarization resistance.[12] The S/MWNTs show a high polarization resistance of 17.9 Ω, whereas the S/MWNT-P25 and S/MWNT-OH25 show lower resistance values of 10.1 Ω and 9.9 Ω, respectively. The similar values of Ohmic resistance but different values of polarization resistance are ascribed to the effect of conductive MWNTs. Fig. 3c provides the cycling performance of each electrode. The initial specific capacities of S/MWNT-P25 and S/MWNT-OH25 increase to 1230 and 1274 mA h g^{-1}, respectively. In contrast, the specific capacity of S/MWNT-OH25 at 100 cycles exhibits a higher cycle retention of 56.6% than that of 49.4% for S/MWNT-P25. Fig. 3d shows the rate retention of each electrode with an increase in the C-rate from 0.1 C to 2 C. The S/MWNT-OH25 exhibits a higher capacity retention of 60% at 2 C than that of S/MWNT and S/MWNT-P25. This performance is attributed to the (1) homogenous distribution of the conductive OH-MWNTs due to a favorable dispersibility of the OH group and (2) hydrophilic

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attraction by the OH-MWNTs. These two properties can provide a continuous conducting pathway inside the sulfur cathode and meanwhile suppress the polysulfide shuttle effect.[13] Even though the Nyquist plots of S/MWNT-P25 and S/MWNT-OH25 show almost the same trend, the cycling performance and rate retention exhibit different electrochemical behavior. The discharge curves of the S/MWNTs with different amounts of hydroxylated CNTs are shown in Fig. S3(a–c) (ESI†). S/MWNT-OH50 exhibited the highest capacity retention in 100 cycles. The cycling performances of these three samples are presented in Fig. S2d (ESI†); the initial discharge specific capacity of S/MWNT-OH50 is 945 mA h g^{-1} and it retained 637 mA h g^{-1} after 100 cycles with a retention of 67.5%, showing higher cycle retention than S/MWNT and S/MWNT-OH25 but relatively lower discharge capacity.

To validate the substantial effect of the OH-group on capturing the polysulfides within the electrode system, we compare the relative quantity of polysulfides in the electrolyte by conducting ultraviolet-visible spectrophotometry (UV-Vis) absorption measurements. As lithium polysulfide strongly absorbs UV radiation, the contents of polysulfide can be demonstrated by the relative intensity of UV absorption.[14] In Fig. 4a, the range of wavelength that correlates with the reaction with bi-products of sulfur and UV light is approximately between 260 and 300 nm. The wavelength of S_2^{2-}, elemental sulfur, and S_6^{2-} are 265 nm, 280 nm and 300 nm, respectively. All of these sulfur species were detected in the electrolyte because they could not be returned back to the original state of sulfur–MWNTs due to the irreversible electrochemical reaction and subsidiary phenomena such as the shuttle effect. With the increased amount of OH-MWNTs, the intensity of the overall absorption peaks relative to the concentration of lithium polysulfides and elemental sulfur in electrolyte decrease distinctly. Thermal gravimetric analysis (TGA) was conducted to examine the content of sulfur within the electrode before and after 100 cycles (Fig. S4, ESI†). The decomposition temperature of PVdF is around 375 °C. The sulfur loading percentages of each of the S/MWNT, S/MWNT-OH25, and S/MWNT-OH50 electrodes before battery testing are 46.4%, 35.8% and 25.1%, respectively. After completing 100 cycles, the content of the remaining sulfur in each of the S/MWNT, S/MWNT-OH25, and S/MWNT-OH50 electrodes was calculated before the temperature reached 375 °C, at which PVdF decomposes, and found to be 20.4%, 19.3%, and 17%, respectively. The percentage of remaining sulfur in the S/MWNT after 100 cycles is almost similar to the numerical value of cycle retention after 100 cycles. After finishing the cycle evaluation, the percentages of remaining sulfur in the S/MWNT, S/MWNT-OH25, and S/MWNT-OH50 electrodes after 100 cycles are 44%, 53.9% and 67.6% (Fig. 4b). The capacity retention values of S/MWNT, S/MWNT-OH25, and S/MWNT-OH50 are 44.5%, 56.6% and 67.5%, which are consistent with the sulfur remaining in the electrode after the corresponding cycles.

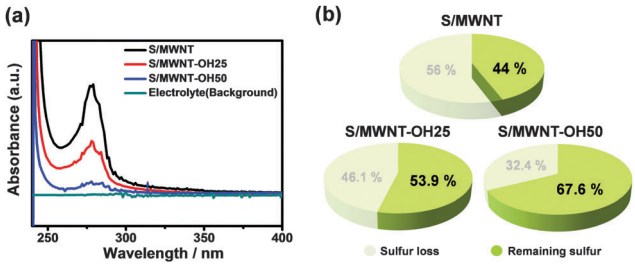

Fig. 4 (a) UV-Vis absorption spectra, (b) chart for the remaining sulfur of the electrodes after 100 cycles of discharge–charge.

In this communication, we introduced hydroxylated-MWNTs into a sulfur cathode in order to diminish the shuttle effect of polysulfides between electrodes in a physical and chemical way and also enhance the conductivity of the sulfur electrode. An *in situ* sulfur deposition method was used to prepare the sulfur–MWNT composite with a core–shell structure, which provides a scalable strategy for making sulfur composites. An increase in the ratio of OH-MWNTs to the sulfur electrode leads to a noticeable enhancement in cycling performance and specific capacity. The S/MWNT-OH25 exhibited a higher specific capacity of 1274 mA h g^{-1} and a more stable cycle retention of 56.6% than those of the S/MWNTs. However, further addition of OH-MWNTs (more than 25 wt%) into the electrode rather inhibits the full activation of sulfur, thereby resulting in a decrease in the specific capacity. The S/MWNT-OH50 showed a lower initial specific capacity of 945 mA h g^{-1} but a higher cycle retention of 67.5% at 100 cycles than those of the S/MWNT-OH50. To clarify this result, the electrochemical performance of the S/MWNT-OH75 is additionally evaluated, exhibiting a much lower specific capacity but stabilized cycling performance, which is plotted in Fig. S6 (ESI†). Extrapolating from these results, the optimal proportion of OH-MWNTs can maximize the specific capacity and improve the cycling performance, whereas excess addition of OH-MWNTs limits the full utilization of sulfur, leading to poor specific capacity. In order to confirm the immediate effect of OH-MWNTs on the electrode after the discharge–charge process, UV-Vis absorption and TGA measurement were conducted. It is proved that the OH-MWNTs are effective in encapturing the dissolved polysulfide within the electrode. We believe that our approach can be further enhanced by modifying the ratio of OH-MWNTs to the sulfur–carbon composite or using the sulfur-impregnated mesoporous carbon to maximize the hydroxyl group effect. In addition, it can be extended to the sulfur composite based on the transition metal oxide (Mn_xO_y, V_xO_y, Ti_xO_y, *etc.*) as well as the carbon material which possesses different hydrophilic oxygen-containing groups.

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